

Effects of On-Site Sanitation on Child Diarrhea and Respiratory Illness: Longitudinal Assessment 1–3.5 Years After Intervention Implementation in a Cluster-Randomized Controlled Trial in Rural Bangladesh

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Objectives

After one to two years of living with the sanitation intervention of the WASH Benefits Bangladesh trial, the seven-day prevalence of diarrheal disease among children under five years old was reduced by 2.2 percentage points compared to children living in control compounds (3.5% vs. 5.7%). At the same time, the seven-day prevalence of acute respiratory illness (ARI) was reduced by 2.5% (6.4% vs. 8.9%). Both reductions were statistically significant. The objective of this analysis is to measure the longer-term effects of the sanitation intervention, which provided latrine upgrades, child feces management tools, and behavioral promotion, on diarrhea and ARI in a random subsample of participants from the same trial. To achieve that objective, we will compare the seven-day prevalence of caregiver-reported diarrheal disease and ARI, measured eight times over a period between 1-3.5 years after the intervention was implemented, among children under five years old living in compounds that received the sanitation intervention vs. children living in control compounds.

Hypotheses

- H1:** Children under five living in compounds in the sanitation arm of the trial will have a lower prevalence of diarrheal disease between 1–3.5 years after intervention implementation compared to households in the control arm.

Rationale: We hypothesize that access to an improved latrine, child potty, sani-scoop, and behavioral messaging would lead to improvements in household fecal waste management, thus reducing transmission of enteric pathogens and preventing diarrheal disease in young children. In the parent trial of this sub-study, the sanitation intervention was found to reduce diarrhea between 1-2 years after implementation (Luby 2018). We hypothesize that this effect will be sustained over an additional 1.5 years.

- H2:** Children under five living in compounds in the sanitation arm of the trial will have a lower prevalence of ARI between 1–3.5 years after intervention implementation compared to households in the control arm.

Rationale: We hypothesize that access to an improved latrine, child potty, sani-scoop, and behavioral messaging would result in improved immune response and better nutritional status of children through fewer enteric infections and cases of diarrhea (Ashraf 2013, 2020; Stewart 2019). Improved immune status and better nutritional status would subsequently result in fewer cases of ARI among young children. In the parent trial of this sub-study, the sanitation intervention was found to reduce ARI between 1-2 years after implementation (Ashraf 2020). We hypothesize that this effect will be sustained over an additional 1.5 years.

- H3:** The effects of the intervention on diarrheal disease and ARI will be greater in the monsoon season compared to the dry season.

Rationale: The prevalence of diarrheal disease among children in rural Bangladesh is higher during the monsoon season than in the dry season, and rainfall after a dry period has been associated with diarrhea (Mertens 2019). The reduction in diarrhea found in the parent trial occurred primarily during the monsoon season (Luby 2018). In tropical climates, the prevalence of ARI is also higher during the rainy season (Murray 2012). We hypothesize that the sanitation intervention will have a larger effect on diarrhea during the monsoon season, when overall environmental contamination is more widespread, and a smaller effect during the dry season, when environmental transmission is reduced. We hypothesize that seasonal differences in intervention effects on ARI would be consistent with effects on diarrhea through immune response and nutritional status.

Study Design

Intervention Trial

The WASH Benefits trial was a randomized controlled trial to measure the effects of water, sanitation, hygiene, and nutrition interventions on child health outcomes in rural Kenya and Bangladesh. Eligible households were those that included a pregnant woman in her first or second trimester who planned to continue living in the village for the next 12 months, with the goal of implementing the intervention before the birth of the child and observing health outcomes during the early years of the child's life. The target household was the household in which the woman who was pregnant during enrollment lived.

Enrolled households were cluster-randomized into study arms, where a cluster consisted of six to eight spatially contiguous households. A buffer zone of at least 15 minutes walking distance (one kilometer) was enforced between each cluster to reduce spillover effects. Eight adjacent clusters were grouped into a study block. Clusters within each block were randomized to one of six intervention arms or into a double-sized control arm, resulting in geographically matched clusters within each block. Geographical matching was performed to account for potential correlation of outcomes by space and to ensure a similar distribution of spatiotemporal variables across study arms.

In the sanitation arm of the trial, the intervention comprised three technologies (a double-pit pour flush improved latrine, a sani-scoop for the removal of feces, and a children's potty) alongside behavioral messaging delivered by trained promoters on the use and maintenance of those technologies. Households in Bangladesh are typically located within multi-family compounds. The sanitation intervention was compound-level, in which the intervention was delivered to all households within target household's compound. Behavioral messaging was continued for the duration of data collection of the parent trial. After enrollment, households in the control arm were not visited by promoters.

The impact of the intervention on diarrheal disease (one of the two primary outcomes of the trial) was assessed at two time points, one and two years after the intervention was implemented. The sanitation intervention was found to reduce the one-week prevalence of diarrheal disease among children under five years old from 5.7% among controls to 3.5% among intervention recipients (PR = 0.61, 95% CI 0.46, 0.81) during this window (Luby 2018). The sanitation intervention also reduced the one-week prevalence of acute respiratory illness (ARI) among children under five from 8.9% among controls to 6.4% among intervention recipients (PR = 0.75, 95% CI 0.58, 0.96) during the same window (Ashraf 2020).

Longitudinal Sub-Study

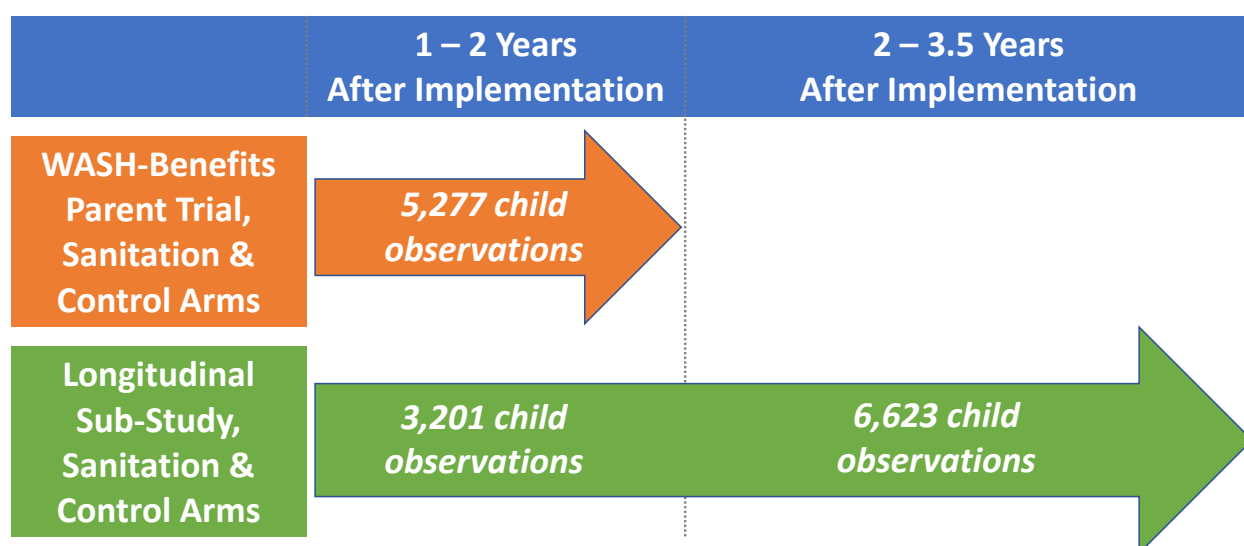
As a sub-study within the parent intervention trial, a longitudinal household survey and environmental assessment was conducted in a randomly selected sample of households from the sanitation and control arms of the Bangladesh trial. Eight households were randomly selected from each study block (four households from the cluster that received the sanitation intervention and four from one of two control clusters). In total, 360 households from the sanitation arm and 360 households from the control arm were selected. Because households were sampled from each cluster, participating households remained pair-matched for the environmental sub-study.

Participating households were visited eight times over 2.5 years. At each visit, participating households completed a survey with questions on children's health including diarrheal symptoms, respiratory symptoms, and negative control outcomes (bruising and abrasions) using the same survey instrument as the parent trial. The mother or primary caregiver of the target child completed the survey and reported symptoms for all children under five present in the compound at the time of the survey.

The sub-study took place between 1–3.5 years after the intervention was implemented, thus providing the opportunity to study the long-term effect of the sanitation intervention on child health beyond the measurements of the parent trial and to compare the consistency of inference between a design that measures outcomes in a larger number of households two times (parent

trial) versus a smaller number of households eight times (the sub-study) (Figure 1). Additionally, observations collected in the sub-study were well distributed between seasons (60% in the monsoon season and 40% in the dry season), allowing us to assess whether intervention effects differ by season.

Figure 1: Time of data collection in relation to implementation of the sanitation intervention for the WASH-Benefits parents trial and this longitudinal sub-study. Data were collected for both studies between 1–2 years after intervention implementation, and data collection for this sub-study continued for 1.5 additional years. A child observation is defined as one survey response for an individual child under five years old; totals include multiple observations of the same individual child.



Data Analysis

Outcomes

The primary outcomes of this analysis are caregiver-reported diarrheal disease and ARI in the past seven days among children under five living in study compounds. A case of diarrheal disease will be defined as a child meeting at least one of two criteria: i) passing three or more stools in 24 hours within the previous seven days and passing watery or loose stools within the previous seven days, or ii) having one or more stool with blood during the previous seven days (Luby 2018; WHO 2019). A case of ARI will be defined as persistent cough, panting, wheezing, or difficulty breathing occurring within the past seven days (Ashraf 2020).

Parameters

The parameters of interest for this analysis are prevalence ratios and prevalence difference of diarrheal disease and ARI comparing children living in compounds in the sanitation arm to those living in compounds in the control arm.

Estimation Strategy

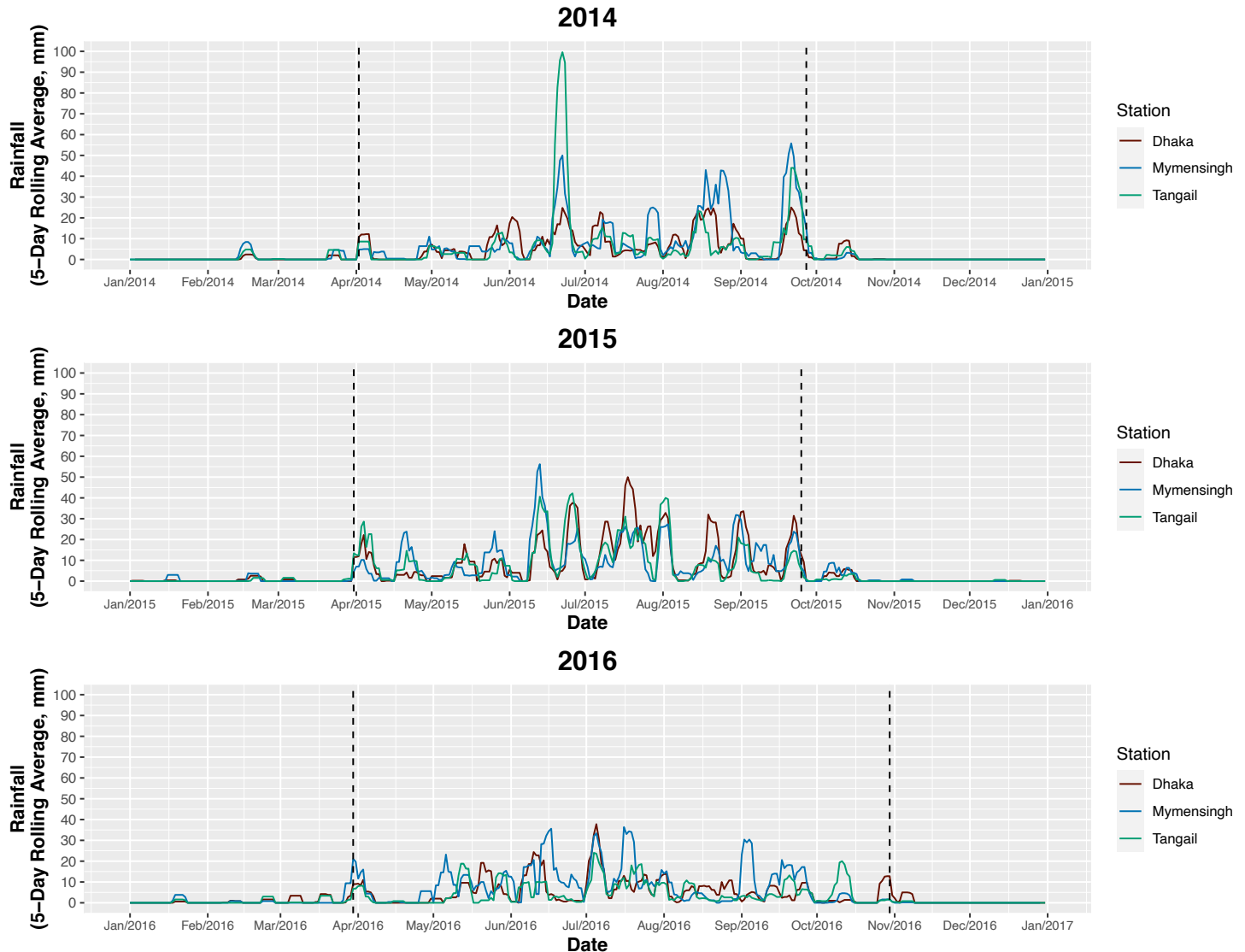
Observations will be pooled across sampling rounds to estimate one prevalence ratio and one prevalence difference per outcome. We will estimate parameters of interest through intention-to-treat analysis using generalized linear models. We will conduct analyses using the *washb* package in R (Mertens and Arnold 2018). To account for cluster-randomization, clustering of children in compounds, and repeated measures of the same children, we will estimate robust sandwich standard errors using study block as the unit of clustering. Matching by geographical location will be accounted for by including study block as a covariate in regression models. Randomization led to good balance of baseline covariates between arms among the subset of households enrolled in the sub-study (Contreras *in prep.*). Therefore, we will not adjust for any additional covariates. Prevalence ratios will be estimated with a log link (family=binomial). Prevalence differences will be estimated with an identity link (family=gaussian). Data analysis scripts will be written using datasets that include randomly assigned values for study arm; true study arm assignment will be unmasked after analysis scripts are complete.

Subgroup Analysis

Subgroup analysis will be conducted to assess differences in intervention effects by season. We will separately estimate intervention effects during the monsoon and dry season for this analysis. We defined the monsoon season using daily rainfall data recorded by the Bangladesh Meteorological Department at three weather stations for the years 2014-2016 (Zaman 2018). Of four districts that make up the study area, two had available weather stations (Mymensingh and Tangail) and two did not (Gazipur and Kishoreganj). We included both available stations, as well as an additional station from Dhaka, which borders the Gazipur district. We calculated five-day

rolling averages of daily rainfall at each station and defined the start of the monsoon season for each year as the first day with a five-day rolling average of 10 mm or greater at any station; we defined the end of the monsoon season for each year as the last day with a five-day rolling average of 10 mm or greater at any station. This definition allowed us to capture the vast majority of rainfall for each year while maintaining a consistent definition across years (Figure 2). Under this definition, the monsoon season for each year was April 2nd – September 27th (2014), March 31st – September 25th (2015), and March 30th – October 30th (2016) (Figure 2).

Figure 2: Daily rainfall (five-day rolling average) at three weather stations in Bangladesh, 2014-2016. Cutoffs for the monsoon season of each year are identified with vertical dashed lines (defined as the first and last days with 10 mm or greater rolling average rainfall at any station).



Sensitivity Analysis

Respondents were asked to report symptoms for all children in the family compound, including their own children and those from other households. It is possible that a respondent more accurately reported symptoms for her own children than for others. We expect any such bias to be non-differential by study arm. As a sensitivity analysis, we will separately estimate effects using observations of the respondents' children and other children in the compound.

Negative Control Analysis

To assess possible bias in outcome measurement, we will measure intervention effects on bruising and abrasions (defined as a child having bruising or abrasions within the previous seven days) using the same methods as for the primary outcomes. These negative control outcomes are not expected to be associated with the sanitation intervention. Therefore, any differences in the prevalence of bruising and abrasions by study arm could indicate bias in our primary analyses.

Missing Data

We previously assessed missingness in the longitudinal sub-study due to loss to follow-up or household availability during survey visits (Contreras *in prep*). We found that the rate of missingness was similar between study arms; we also found that enrollment characteristics were similar among households that completed all surveys compared to households that were unavailable during one or more survey rounds (Contreras *in prep*). Based on these findings, we will assume outcomes are missing completely at random, and will include all observations measured in this study.

Minimum Detectable Effects

We recorded 9,819 observations of diarrheal disease from children under five years old. Among 4,722 observations in the control arm, the seven-day prevalence of diarrheal disease was 14.5%. There was an average of 54.6 observations per cluster in each study arm. The average coefficient of variation of cluster sizes was 0.30. Using the analysis of variance method of estimation, the intraclass correlation coefficient for diarrheal disease by study cluster was 0.04.

These values were used to estimate the prevalence of diarrheal disease in the sanitation arm that we are able to detect with 80% power and a two-sided alpha of 0.05 through the clusterPower package in R (Moyer and Kleinman 2021). The available observations yield 80% power to detect a prevalence ratio for diarrheal disease of 0.77 and a prevalence difference of -3.3 percentage points in the sanitation arm vs. control arm (11.2% estimated prevalence in sanitation arm compared to 14.5% observed prevalence in the control arm).

We recorded 9,824 observations of ARI from children under five years old. Among 4,725 observations in the control arm, the seven-day prevalence of ARI was 22.6%. Other values were the same as those for diarrheal disease. With an alpha of 0.05, available observations yield 80% power to detect prevalence ratio for ARI of 0.84 and a prevalence difference of -3.5 percentage points in the sanitation arm vs. control arm (19.1% estimated prevalence in sanitation arm compared to 22.6% observed prevalence in the control arm).

We used the same methods to estimate power based on the available sample size during the monsoon and dry seasons, separately. During the monsoon season, the available observations yield 80% power to detect a prevalence ratio for diarrheal disease of 0.74 and a prevalence difference of -3.9 percentage points (11.4% estimated prevalence in the sanitation arm compared to 15.3% observed prevalence in the control arm) and 80% power to detect a prevalence ratio for ARI of 0.80 and a prevalence difference of -4.0 percentage points (16.1% estimated prevalence in the sanitation arm compared to 20.0% observed prevalence in the control arm). During the dry season, the available observations yield 80% power to detect a

prevalence ratio for diarrheal disease of 0.71 and a prevalence difference of –3.8 percentage points (9.4% estimated prevalence in the sanitation arm compared to 13.2% observed prevalence in the control arm) and 80% power to detect a prevalence ratio for ARI of 0.81 and a prevalence difference of –5.1 percentage points (21.4% estimated prevalence in the sanitation arm compared to 26.5% observed prevalence in the control arm).

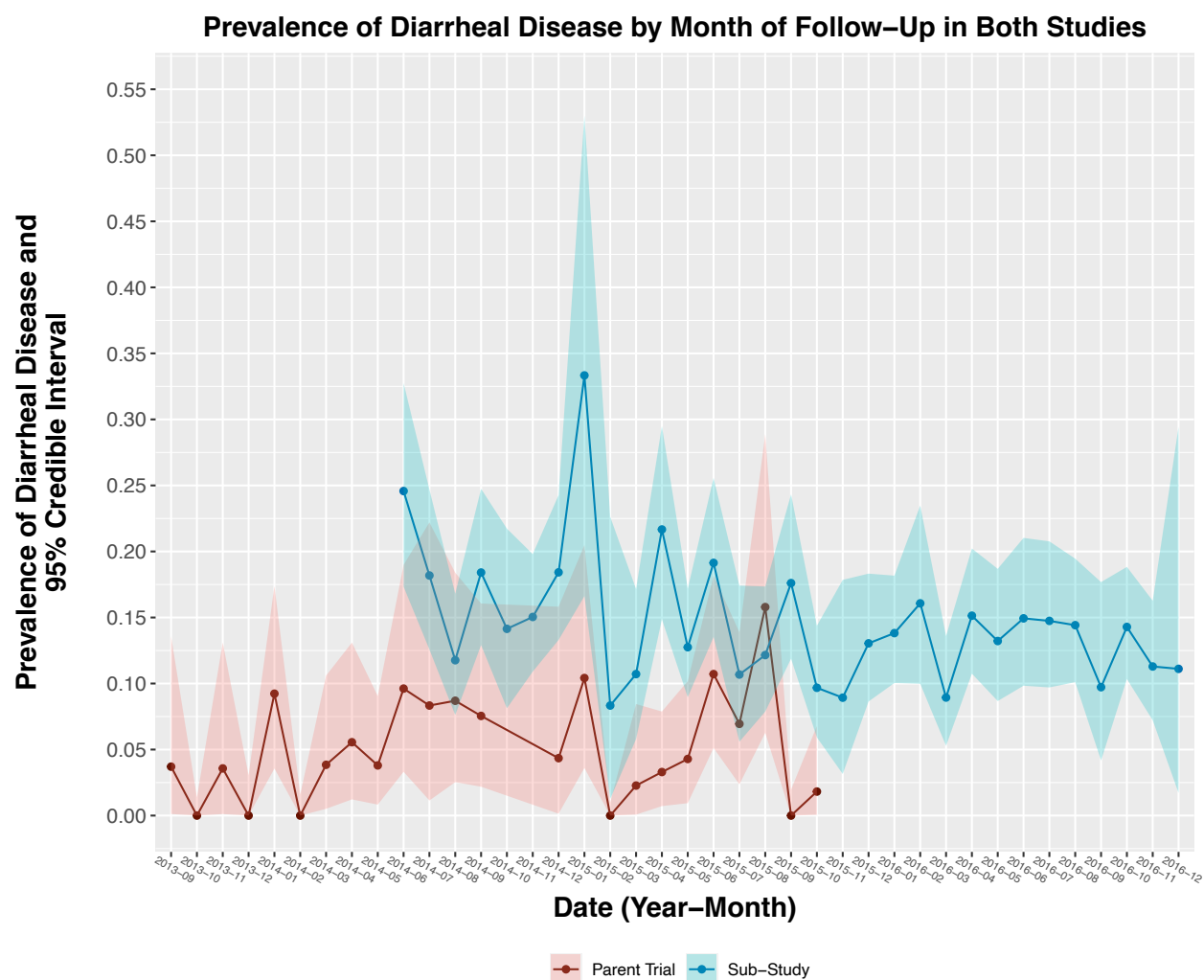
Outcome Prevalence in Parent Trial vs. Longitudinal Sub-Study

While estimating minimal detectable effects, we observed a notably higher prevalence of diarrheal disease and ARI among controls in this sub-study (14.5% diarrheal disease; 22.6% ARI) compared to prevalence measured in the full control group by the parent trial (5.7% diarrheal disease; 8.9% ARI) (Luby 2018). We investigated whether this difference is due to selection of the random sample of participants that were enrolled in the sub-study or due to temporal differences between the data collection periods.

First, we assessed whether the sample of households that participated in the sub-study experienced a higher prevalence of disease in the parent trial compared to other households in the control group, possibly due to random variation in the sample, or whether the higher prevalence was unique to the sub-study. We calculated the prevalence of diarrheal disease observed in the parent trial among control compounds that participated in the sub-study (5.4%) and found little variation compared to the prevalence measured by the parent trial in the control group as a whole (5.7%).

Next, we measured the prevalence of diarrheal disease among the same compounds by month of follow-up time to determine if the discrepancy was related to temporal trends (Figure 3). The prevalence reported in the sub-study was generally higher than that reported in the parent trial even during overlapping follow-up periods. Ninety-five percent credible intervals overlapped between studies for most overlapping months, likely due to small sample size in individual months. We did not find evidence that the discrepancy in diarrheal prevalence between studies was due to temporal trends.

Figure 3: Prevalence of diarrheal prevalence among children under five years old by month of follow-up in the WASH Benefits parent trial and longitudinal sub-study. All data were restricted to compounds in the control arm that participated in the longitudinal sub-study (n compounds = 360). The overall prevalence among all observations within each study (5.4% for the parent trial, 14.5% for the sub-study) was used as a prior distribution to estimate 95% credible intervals around prevalence values (Brown 2001).



Last, we compared the prevalence of bruising or abrasion between the control groups of each study to test if the difference in outcome prevalence between studies also held true for a negative control outcome that is unrelated to the objectives of the trial. The prevalence of bruising or abrasion among control children in the sub-study (8.9%) was higher than the prevalence among the overall control group of the parent trial (2.9%), suggesting that the differences in outcome prevalence found between studies is consistent for all child health outcomes.

We conclude that the outcome assessment methods employed in the sub-study captured a higher prevalence of child health outcomes compared to the parent trial. Survey questions to measure these outcomes, survey methodology protocols, and definition and operationalization of outcome variables were identical between the studies. We note that an earlier environmental assessment study, conducted in July 2013 through March 2014, captured a diarrheal prevalence of 17–18% among children under five years old using the same survey questions, consistent with the prevalence in this sub-study (Pickering 2018). It is possible that, because these studies primarily focused on environmental sampling rather than health outcomes, they elicited a more honest response from caregivers on child health symptoms. In the parent trial, where health symptoms were a primary focus, health outcomes may have been underreported due to courtesy bias or social desirability bias. If there was measurement error in this sub-study, and if that error was differential by intervention arm, estimates of intervention effects on child health outcomes could be biased. We will assess potential bias in our primary estimates by estimating the effect of the intervention on bruising and abrasion (see Negative Control Analysis above). Bruising and abrasion was more prevalent in this sub-study but was not reasonably affected by the intervention. Thus, finding no effect of the intervention on bruising and abrasion would suggest that any potential measurement error is less likely to be differential by intervention arm and that primary effect estimates are unbiased.

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